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* Find the dual of the following primal problem.

① $\text{Max } Z = 40x_1 + 35x_2$

s.t. $2x_1 + 3x_2 \leq 60$

$4x_1 + 3x_2 \leq 96$

& $x_1, x_2 \geq 0$

→ Dual is given by

$\text{Min } Z_w = 60w_1 + 96w_2$

s.t. $2w_1 + 4w_2 \geq 40$

$3w_1 + 3w_2 \geq 35$

& $w_1, w_2 \geq 0$

② $\text{Min } Z = 10x_1 + 20x_2$

s.t. $3x_1 + 2x_2 \geq 18$

$x_1 + 3x_2 \geq 8$

$2x_1 - 5x_2 \leq 6$

& $x_1, x_2 \geq 0$

→ The given LPP convert in standard form

$\text{Min } Z = 10x_1 + 20x_2$

s.t. $3x_1 + 2x_2 \geq 18$

$x_1 + 3x_2 \geq 8$

$-2x_1 + x_2 \geq -6$ & $x_1, x_2 \geq 0$

Now, its dual is given by

$\text{Max } Z_y = 18y_1 + 8y_2 - 6y_3$

s.t. $3y_1 + y_2 - 2y_3 \leq 10$

$2y_1 + 3y_2 + y_3 \leq 20$

& $y_1, y_2 \geq 0$

③ Max $Z_x = 5x_1 + 12x_2 + 4x_3$

s.t. $x_1 + 2x_2 + x_3 \leq 5$

$2x_1 - x_2 + 3x_3 \geq 2$

$x_1, x_2, x_3 \geq 0$

also verify that dual of dual is primal

→ The standard primal of the given primal is.

Max $Z_x = 5x_1 + 12x_2 + 4x_3$

s.t. $x_1 + 2x_2 + x_3 \leq 5$

$2x_1 - x_2 + 3x_3 \leq 2$

$-2x_1 + x_2 - 3x_3 \leq -2$

$x_1, x_2, x_3 \geq 0$

Now, its dual is

Min $Z_w = 5w_1 + 2(w_2' - w_2'')$

s.t. $w_1 + 2(w_2' - w_2'') \geq 5$

$2w_1 - (w_2' + w_2'') \geq 12$

$w_1 + 3(w_2' - w_2'') \geq 4$

$w_1, w_2', w_2'' \geq 0$

put $w_2 = (w_2' - w_2'')$

then above dual takes the following form.

min $Z_w = 5w_1 + 2w_2$

s.t. $w_1 + 2w_2 \geq 5$

$2w_1 - w_2 \geq 12$

$w_1 + 3w_2 \geq 4$

$w_1 \geq 0$, w_2 is unrestricted in sign.

Dual of dual is

Max $Z_y = 5y_1 + 12y_2 + 4y_3$

s.t. $y_1 + 2y_2 + y_3 \leq 5$

$2y_1 - y_2 + 3y_3 \leq 2$

$y_1, y_2, y_3 \geq 0$

∴ Dual of dual is primal